Tutorial 8

Linear independence, basis and dimension

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1. *This question asks you to try to understand what linear dependence means geometrically, in R^3. Try to visualize the sets of vectors.*
For each of the following sets of vectors in $\mathbb{R}^3$, say whether it is
(i) always (ii) never (iii) sometimes
linearly independent.
If you choose (iii), say when the vectors will be linearly independent.
   (a) Two distinct (different) non-zero vectors on a plane through the origin.

   (b) Three distinct non-zero vectors on a plane through the origin.

   (c) Two distinct non-zero vectors in $\mathbb{R}^3$.

   (d) Three distinct non-zero vectors in $\mathbb{R}^3$.

2. Let S and T be subsets of $M_{2\times 2}(\mathbb{R})$ defined by

$$S = \left\{ \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix} \right\} \text{ and } T = \left\{ \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix} \right\}.$$

In Question 6 in Tutorial 7 you found that

$$\text{span } S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2\times 2}(\mathbb{R}) : a + b - c - d = 0 \right\},$$

and that it is possible to find a 2×2 matrix which is not in span(S).

   (a) Is S a linearly independent subset of $M_{2\times 2}(\mathbb{R})$? Give a reason for your answer.

   Try to use your work in Question 6 of Tutorial 7 to answer the following questions quickly.

   (b) Is T a linearly independent subset of $M_{2\times 2}(\mathbb{R})$? Give a reason for your answer; you should be able to do this without doing any calculations.

   (c) Find span T. How many equations will be needed to describe span T? Find the equation or equations. Write your answer in the same form as that for span(S) above. You should be able to use your working for Question 6 of Tutorial 7.

3. Without doing any lengthy calculations, state whether or not the following subsets of vectors form a basis for the given vector space. Give reasons for your answers. In this question you can use any of the useful facts we have covered in lectures concerning bases. This should go quickly; use your notes!

   (a) $\left\{ \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 5 & 6 \end{pmatrix}, \begin{pmatrix} 9 & 2 \\ 1 & 4 \end{pmatrix} \right\}$ in $M_{2\times 2}(\mathbb{R})$.

   (b) $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$ in $\mathbb{R}^3$.

   (c) $\{x^2 + x,\ 3x,\ 4x^2 + 1,\ -17\}$ in P_2.

   (d) $\{x,\ x - 1,\ x^2 - x\}$ in P_2.

   (e) $\left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 4 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ in $\mathbb{R}^4$.

4. Let $B = \{x^2 + 1,\ x - 1,\ x^2 - x\}$.

   (a) Show that B is a basis for P_2.

   (b) Find the coordinate vector of $5x^2 - 9$ with respect to the basis B.

5. Read the following result and its proof, and then answer the questions following it.

   Theorem Let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ be a linearly independent set in a vector space V. If $\boldsymbol{v}$ is a vector in V that is not in the span of S then $S \cup \{\boldsymbol{v}\}$ is still linearly independent.

   Proof. Suppose that $\boldsymbol{v} \notin \text{span}(S)$, but $S \cup \{\boldsymbol{v}\}$ is linearly dependent. Then there are scalars $\lambda_1, \lambda_2, \ldots, \lambda_{n+1}$, not all zero, such that

   $$\lambda_1 \boldsymbol{v}_1 + \lambda_2 \boldsymbol{v}_2 + \ldots + \lambda_n \boldsymbol{v}_n + \lambda_{n+1} \boldsymbol{v} = \mathbf{0}.$$

   We have $\lambda_{n+1} \neq 0$, since S is linearly independent. Then

   $$\boldsymbol{v} = -\frac{1}{\lambda_{n+1}} [\lambda_1 \boldsymbol{v}_1 + \lambda_2 \boldsymbol{v}_2 + \ldots + \lambda_n \boldsymbol{v}_n].$$

   But this is a contradiction, so $S \cup \{\boldsymbol{v}\}$ must be linearly independent.

   □

(a) What method of proof is used to prove this result?

(b) How does it follow from the linear independence of S that $\lambda_{n+1} \neq 0$?

(c) What is the contradiction referred to in the last line of the proof?

(d) Let S be the set defined in Question 2. Is there a set B **containing** S which has four matrices in it and is independent? If so, find one; if not, explain why not. (If you have understood the previous questions, this should require no calculation at all.)

(e) Will the theorem still be true if we replace "If $\boldsymbol{v}$ is a vector in V that is not in the span of S then ..." by "If $\boldsymbol{v}$ is a vector in V that is not in S then ..."? Give a reason for your answer.